

Agenda

- ⬡ What is RDBMS.
- ⬡ Tables, Columns and Data types
- ⬡ How to design a database.
- ⬡ One-To-Many & Many-To-Many Relationships.
- ⬡ Activity: FB, YouTube, Talabat
- ⬡ SQL
- ⬡ CRUD
- ⬡ Selecting data
- ⬡ Filtering data
- ⬡ Ordering data
- ⬡ Limiting data

- ⬡ Aggregate Functions
- ⬡ Grouping data
- ⬡ Datetime Operations
- ⬡ Joining tables
- ⬡ Subqueries
- ⬡ Window Functions
- ⬡ Inserting new data
- ⬡ Updating data
- ⬡ Deleting data
- ⬡ Python & MySQL
- ⬡ MySQL Workbench

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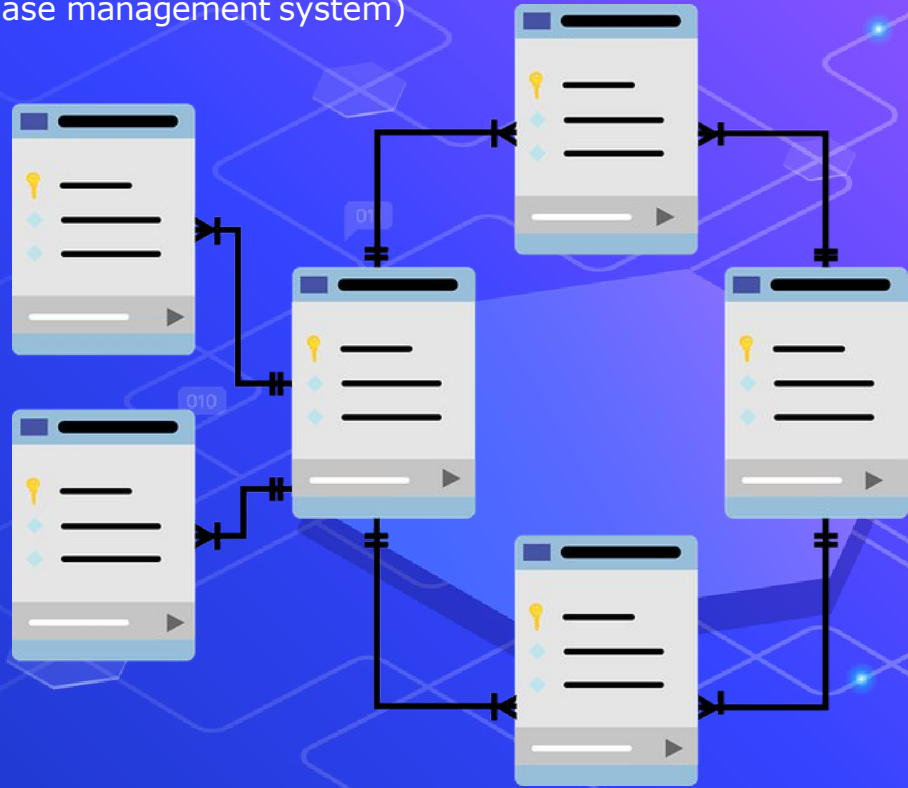
What is RDBMS. (relational database management system)

File systems made for storing and manipulating relational data.

Data organized in tabular form each table represents an entity like users, products, categories, etc...

Each table consists of columns that represent entity properties, each column can be any data type like numbers, text, etc...

SQL is an easy language to connect with database and query the data from it easily.



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Tables, Columns and Data types

Tables

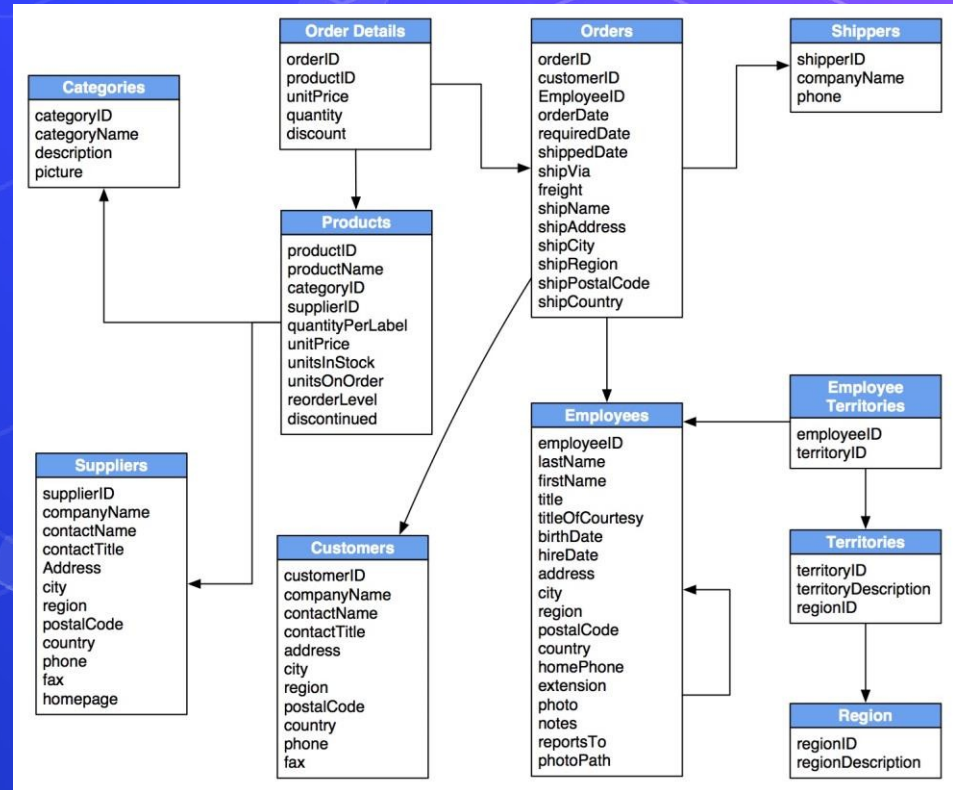
Users, posts, comments, products, ...

Columns

Users table have columns like age, gender, username, email, is_online,...

Data Types

Age is Numeric, username is TEXT
Gender is CHAR, is_online is BOOL



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How to design a database.

1) Analyze the system and divide it to its main entities (tables).

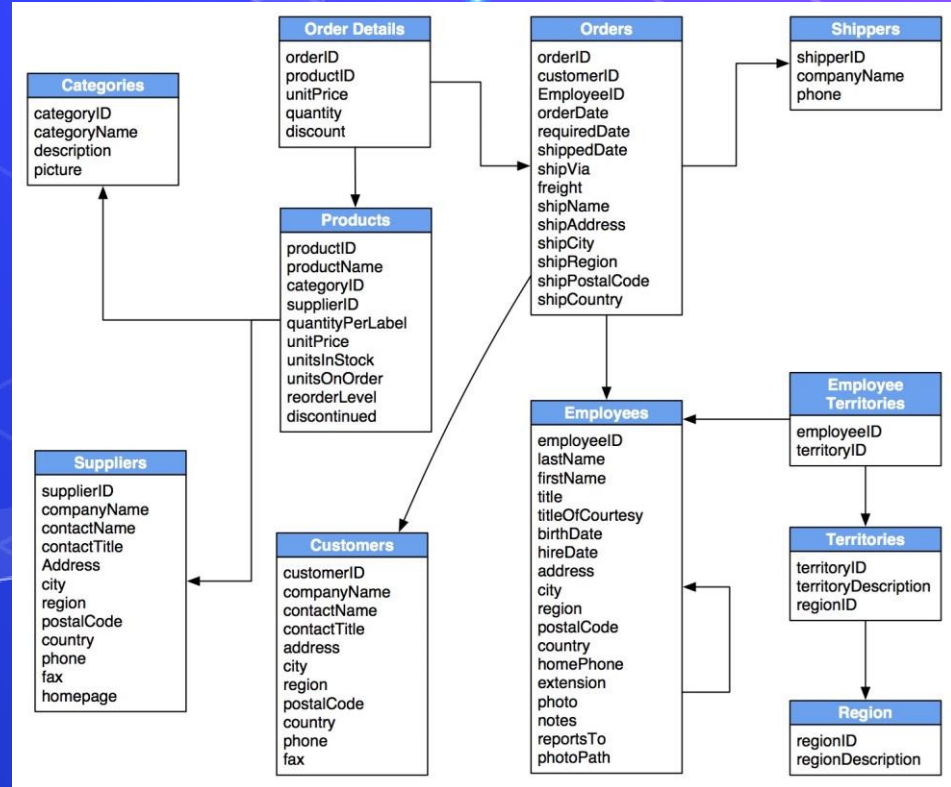
for example we make an ecommerce system so the main entities tables are Categories, Products, Orders, Customers, etc...

2) For each table define what its properties (columns) are and what is their data types.

for example customers table have name, city, phone and products table have price, name, etc..., the most important thing don't forget to add the unique auto increment **id** column (Primary Key) because it's the row identifier.

3) Define direct tables relationships and based on the relationship define the foreign keys.

for example one-to-many relationship between the products and categories tables so we define the **CategoryID** column (Foreign Key) in the products table, another relation is many-to-many relation between orders and products tables so we make Order Details table that contain the foreign keys



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One-To-Many & Many-To-Many Relationships

Products --->

CategoryID is a foreign key that points to the CategoryID primary key in categories table

ProductID	ProductName	SupplierID	CategoryID	Unit	Price
1	Chais	1	1	10 boxes x 20 bags	18
2	Chang	1	1	24 - 12 oz bottles	19
3	Aniseed Syrup	1	2	12 - 550 ml bottles	10
4	Chef Anton's Cajun Seasoning	2	2	48 - 6 oz jars	22
5	Chef Anton's Gumbo Mix	2	2	36 boxes	21.35
6	Grandma's Boysenberry Spread	3	2	12 - 8 oz jars	25
7	Uncle Bob's Organic Dried Pears	3	7	12 - 1 lb pkgs.	30
8	Northwoods Cranberry Sauce	3	2	12 - 12 oz jars	40
9	Mishi Kobe Niku	4	6	18 - 500 g pkgs.	97
10	Ikura	4	8	12 - 200 ml jars	31

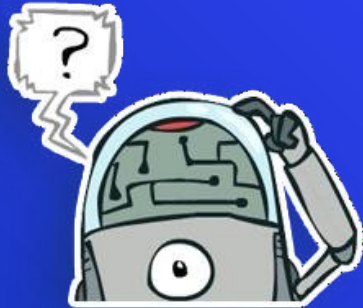
CategoryID	CategoryName	Description
1	Beverages	Soft drinks, coffees, teas, beers, and ales
2	Condiments	Sweet and savory sauces, relishes, spreads, and seasonings
3	Confections	Desserts, candies, and sweet breads
4	Dairy Products	Cheeses
5	Grains/Cereals	Breads, crackers, pasta, and cereal
6	Meat/Poultry	Prepared meats
7	Produce	Dried fruit and bean curd
8	Seafood	Seaweed and fish

<--- Categories

CategoryID is a primary key

One-To-Many & Many-To-Many Relationships

Products --->



How can we
represent the
relationship ?!

Orders --->

ProductID	ProductName	SupplierID	CategoryID	Unit
1	Chais	1	1	10 box
2	Chang	1	1	24 - 12
3	Aniseed Syrup	1	2	12 - 55
4	Chef Anton's Cajun Seasoning	2	2	48 - 6
5	Chef Anton's Gumbo Mix	2	2	36 box
6	Grandma's Boysenberry Spread	3	2	12 - 8
7	Uncle Bob's Organic Dried Pears	3	7	12 - 1
8	Northwoods Cranberry Sauce	3	2	12 - 12
9	Mishi Kobe Niku	4	6	18 - 50
10	Ikura	4	8	12 - 20

OrderID	CustomerID	EmployeeID	OrderDate
10248	90	5	1996-07-04
10249	81	6	1996-07-05
10250	34	4	1996-07-08
10251	84	3	1996-07-08
10252	76	4	1996-07-09
10253	34	3	1996-07-10
10254	14	5	1996-07-11
10255	68	9	1996-07-12

One-To-Many & Many-To-Many Relationships

Products

ProductID	ProductName
1	Chais
2	Chang
3	Aniseed Syrup
4	Chef Anton's Cajun Seasoning
5	Chef Anton's Gumbo Mix
6	Grandma's Boysenberry Spread
7	Uncle Bob's Organic Dried Pears
8	Northwoods Cranberry Sauce
9	Mishi Kobe Niku
10	Ikura

Order Details

OrderDetailID	OrderID	ProductID	Quantity
1	10248	1	12
2	10248	42	10
3	10248	72	5
4	10249	14	9
5	10249	51	40
6	10250	41	10
7	10250	1	35
8	10250	65	15
9	10251	1	6
10	10251	57	15

Orders

OrderID	CustomerID
10248	90
10249	81
10250	34
10251	84
10252	76
10253	34
10254	14

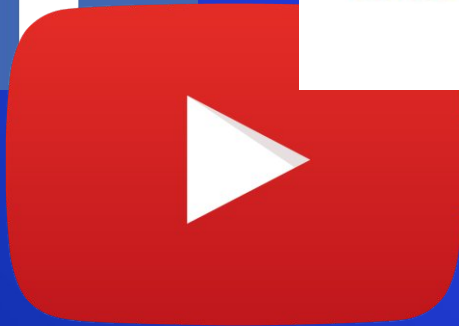
We make a new table that have both foreign keys

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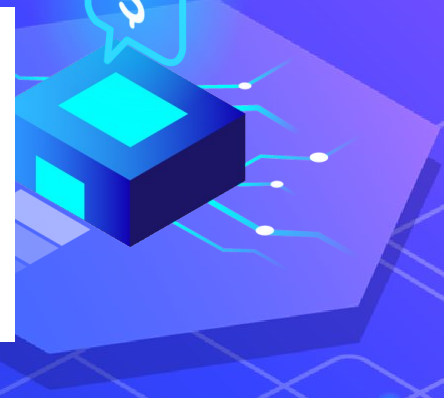
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Examples: FB, Youtube, Talabat



talabat



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SQL

SQL or Structured Query Language used to communicate with the database by executing commands for creating tables and columns and also reading, filtering, ordering, inserting, updating, deleting data and more.

You can try SQL on bunch of examples from this link

<https://www.w3schools.com/sql/default.asp>

<https://www.sqlitetutorial.net>



SQL Engines

SQL Engines/Servers are programs that host an RDBMS system where you can store databases and perform read and write operations to it

There are a variety of options for SQL hosting. The most famous ones are:

- ❖ **MySQL**
- ❖ Microsoft SQL Server
- ❖ PostgreSQL
- ❖ SQLite (Offline SQL)
- ❖ ...



MySQL

MySQL is one of the most used SQL servers
It's very flexible and offers a lot of useful tools and features

```
mysql> SELECT film_id, title, length FROM film LIMIT 50;
```

film_id	title	length
1	ACADEMY DINOSAUR	86
2	ACE GOLDFINGER	48
3	ADAPTATION HOLES	50
4	AFFAIR PREJUDICE	117
5	AFRICAN EGG	130
6	AGENT TRUMAN	169
7	AIRPLANE SIERRA	62
8	AIRPORT POLLOCK	54
9	ALABAMA DEVIL	114
10	ALADDIN CALENDAR	63
11	ALAMO VIDEOTAPE	126
12	ALASKA PHANTOM	136
13	ALI FOREVER	150
14	ALICE FANTASIA	94
15	ALIEN CENTER	46
16	ALLEY EVOLUTION	180
17	ALONE TRIP	82
18	ALTER VICTORY	57
19	AMADEUS HOLY	113
20	AMELIE HELLFIGHTERS	79
21	AMERICAN CIRCUS	129

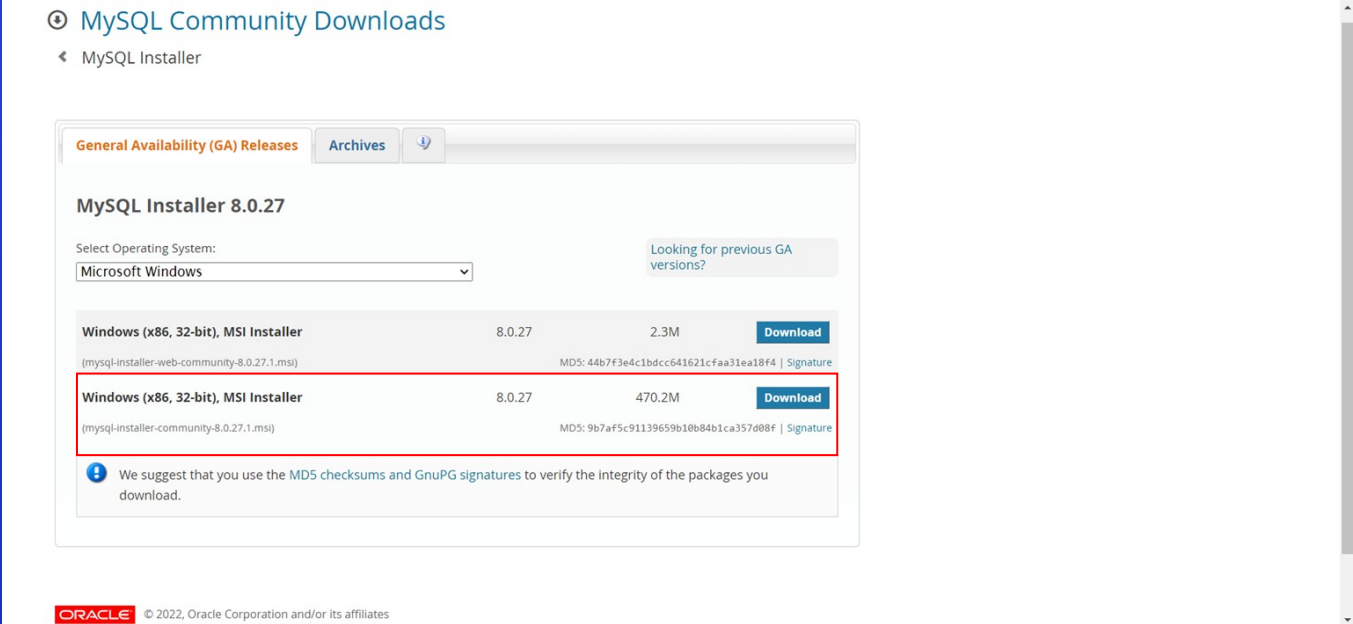
The screenshot shows the MySQL Workbench interface. On the left, the 'SCHEMAS' pane displays the 'sakila' database structure, including tables like actor, address, city, country, customer, film, film_actor, film_category, film_text, inventory, language, payment, rental, staff, and store. The 'Tables' pane shows the 'film' table structure with columns: film_id, title, description, release_year, language_id, original_language_id, rental_duration, rental_rate, length, replacement_cost, rating, and special_features. The main query editor shows the query: `SELECT * FROM sakila.film;`. The 'Result Grid' pane displays the query results, showing a list of films with their details. The 'Output' pane at the bottom shows the query execution time and duration.

film_id	title	description	release_year	language_id	original_language_id	rental_duration	rental_rate	length	replacement_cost	rating	special_features
1	ACADEMY DINOSAUR	An Epic Drama of a Feminist And a Mad Scientist ...	2005	1	1	6	0.99	86	20.99	PG	Deleted i...
2	ACE GOLDFINGER	A Astounding Epistle of a Database Administrat...	2006	1	1	3	4.99	48	12.99	G	Trailers,I
3	ADAPTATION HOLES	A Astounding Reflection of a Lumberjack And a ...	2006	1	1	7	2.99	50	18.99	NC-17	Trailers,I
4	AFFAIR PREJUDICE	A Fanciful Documentary of a Frisbee And a Lum...	2006	1	1	5	2.99	117	26.99	G	Comment
5	AFRICAN EGG	A Fast-Paced Documentary of a Pastry Chef An...	2006	1	1	6	2.99	130	22.99	G	Deleted i...
6	AGENT TRUMAN	A Intrepid Panorama of a Robot And a Boy who...	2006	1	1	3	2.99	169	17.99	PG	Deleted i...
7	AIRPLANE SIERRA	A Touching Saga of a Hunter And a Butler who...	2006	1	1	6	4.99	62	28.99	PG-13	Trailers,I
8	AIRPORT POLLOCK	A Epic Tale of a Moose And a Girl who must Con...	2006	1	1	6	4.99	54	15.99	R	Trailers
9	ALABAMA DEVIL	A Thoughtful Panorama of a Database Administr...	2006	1	1	3	2.99	114	21.99	PG-13	Trailers,I
10	ALADDIN CALENDAR	A Action-Packed Tale of a Man And a Lumberjac...	2006	1	1	6	4.99	63	24.99	NC-17	Trailers,I
11	ALAMO VIDEOTAPE	A Boring Epistle of a Butler And a Cat who must...	2006	1	1	6	0.99	126	16.99	G	Comment
12	ALASKA PHANTOM	A Fanciful Saga of a Hunter And a Pastry Chef...	2006	1	1	6	0.99	136	22.99	PG	Comment

Installing MySQL

You can download MySQL installer for Windows through this link:

<https://dev.mysql.com/downloads/installer/>



The screenshot shows the MySQL Community Downloads page for the MySQL Installer 8.0.27. The page has a blue header with the MySQL logo and the text "MySQL Community Downloads". Below the header, there is a breadcrumb trail: "MySQL Installer". The main content area has a tabbed interface with "General Availability (GA) Releases" selected. The title "MySQL Installer 8.0.27" is displayed. Below the title, there is a "Select Operating System:" dropdown menu set to "Microsoft Windows". To the right of the dropdown is a link "Looking for previous GA versions?". Below this, there is a table of download links. The table has two rows, both for "Windows (x86, 32-bit), MSI Installer". The first row is for version 8.0.27, size 2.3M, and has a "Download" button. The second row is also for version 8.0.27, size 470.2M, and has a "Download" button. The second row is highlighted with a red border. Below the table, there is a blue information icon and a message: "We suggest that you use the MD5 checksums and GnuPG signatures to verify the integrity of the packages you download." At the bottom of the page, there is an Oracle logo and the text "© 2022, Oracle Corporation and/or its affiliates".

MySQL Community Downloads

MySQL Installer

General Availability (GA) Releases Archives

MySQL Installer 8.0.27

Select Operating System:
Microsoft Windows

Looking for previous GA versions?

Windows (x86, 32-bit), MSI Installer	8.0.27	2.3M	Download
(mysql-installer-web-community-8.0.27.1.msi) MD5: 44b7f3e4c1bdcc641621cfaa31ea18f4 Signature			
Windows (x86, 32-bit), MSI Installer	8.0.27	470.2M	Download
(mysql-installer-community-8.0.27.1.msi) MD5: 9b7af5c91139659b10b84b1ca357008f Signature			

We suggest that you use the MD5 checksums and GnuPG signatures to verify the integrity of the packages you download.

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MySQL Data Types

In SQL, each column can have one data type only. MySQL features a number of data types, each with certain properties.

Data Types are split into three categories:

- ❖ Numeric
- ❖ String (Text)
- ❖ Datetime

You can find a list of the most important ones here:
https://www.w3schools.com/sql/sql_datatypes.asp

Numeric Data Types

Table 11.1 Required Storage and Range for Integer Types Supported by MySQL

Type	Storage (Bytes)	Minimum Value Signed	Minimum Value Unsigned	Maximum Value Signed	Maximum Value Unsigned
TINYINT	1	-128	0	127	255
SMALLINT	2	-32768	0	32767	65535
MEDIUMINT	3	-8388608	0	8388607	16777215
INT	4	-2147483648	0	2147483647	4294967295
BIGINT	8	-2^{63}	0	$2^{63}-1$	$2^{64}-1$

There's also:

- ❖ FLOAT
- ❖ DOUBLE
- ❖ DECIMAL

String Data Types

TABLE 4.2 MySQL Data Types

Type	Size	Description
CHAR [Length]	Length bytes	A fixed-length field from 0 to 255 characters long
VARCHAR [Length]	String length + 1 or 2 bytes	A variable-length field from 0 to 65,535 characters long
TINYTEXT	String length + 1 bytes	A string with a maximum length of 255 characters
TEXT	String length + 2 bytes	A string with a maximum length of 65,535 characters
MEDIUMTEXT	String length + 3 bytes	A string with a maximum length of 16,777,215 characters
LONGTEXT	String length + 4 bytes	A string with a maximum length of 4,294,967,295 characters

Datetime Data Types

Data Type	"Zero" Value
<u>DATE</u>	'0000-00-00'
<u>TIME</u>	'00:00:00'
<u>DATETIME</u>	'0000-00-00 00:00:00'
<u>TIMESTAMP</u>	'0000-00-00 00:00:00'
<u>YEAR</u>	0000

See also:

<https://dev.mysql.com/doc/refman/8.0/en/data-types.html>

Project #8 Design Ecommerce DB

W3

schoools



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CRUD

SQL supports these operations:

CRUD

Create

Read

Update

Delete



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Selecting data

- ⬡ Select all columns from a table
- ⬡ Select specific columns from a table



```
1 SELECT * FROM Customers
```



Selecting data

- ⬡ Select all columns from a table
- ⬡ Select specific columns from a table



```
1 SELECT CustomerName, City FROM Customers
```



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Filtering data

WHERE Clause

-  Filtering with Text vs Numeric Fields
-  Comparison operators (>, <, >=, <=, =, <>)
-  BETWEEN
-  IN
-  LIKE
-  Logic operators (AND, OR, NOT)
-  IS NULL, IS NOT NULL



```
1 SELECT * FROM Customers WHERE {STATEMENT}
```

Filtering data

- WHERE Clause
- Filtering with Text vs Numeric Fields**
- Comparison operators (>, <, >=, <=, =, <>)
- BETWEEN
- IN
- LIKE
- Logic operators (AND, OR, NOT)
- IS NULL, IS NOT NULL

```
1 -- With Text Field --
2 SELECT * FROM Customers WHERE Country='Mexico'
3
4
5 -- With Numeric Field --
6 SELECT * FROM Products WHERE Price=30
7
```

Filtering data

- WHERE Clause
- Filtering with Text vs Numeric Fields
- Comparison operators (=, !=, >, <, >=, <=)
- BETWEEN
- IN
- LIKE
- Logic operators (AND, OR, NOT)
- IS NULL, IS NOT NULL

```
1 SELECT * FROM Customers WHERE Country = 'Mexico'
2
3 SELECT * FROM Customers WHERE Age != 25
4
5 SELECT * FROM Products WHERE Price > 30
6
7 SELECT * FROM Books WHERE Rate >= 3
8
9 SELECT * FROM Products WHERE Price < 30
10
11 SELECT * FROM Books WHERE Rate <= 3
```

Filtering data

- WHERE Clause
- Filtering with Text vs Numeric Fields
- Comparison operators (=, !=, >, <, >=, <=)
- BETWEEN**
- IN
- LIKE
- Logic operators (AND, OR, NOT)
- IS NULL, IS NOT NULL

```
1 SELECT * FROM Products
2 WHERE Price BETWEEN 10 AND 20
3
4
5 SELECT * FROM Products
6 WHERE Price NOT BETWEEN 10 AND 20
```

Filtering data

- WHERE Clause
- Filtering with Text vs Numeric Fields
- Comparison operators (=, !=, >, <, >=, <=)
- BETWEEN
- IN
- LIKE
- Logic operators (AND, OR, NOT)
- IS NULL, IS NOT NULL

```
1 SELECT * FROM Customers
2 WHERE Country IN ('Germany', 'France', 'UK')
3
4
5 SELECT * FROM Products
6 WHERE Price NOT IN (10, 30, 50)
```

Filtering data

- WHERE Clause
- Filtering with Text vs Numeric Fields
- Comparison operators (=, !=, >, <, >=, <=)
- BETWEEN
- IN
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```
1 -- Select rows that start with 'a'
2 SELECT * FROM Customers WHERE CustomerName LIKE 'a%'
3
4
5 -- Select rows that start with any letter except 'a'
6 SELECT * FROM Customers WHERE CustomerName NOT LIKE 'a%'
7
8
9 -- Select rows that end with 'a'
10 SELECT * FROM Customers WHERE CustomerName LIKE '%a'
11
12
13 -- Select rows that have 'or' in any position
14 SELECT * FROM Customers WHERE CustomerName LIKE '%or%'
15
16
17 -- Select rows that start with 'a' and end with 'o'
18 SELECT * FROM Customers WHERE CustomerName LIKE 'a%o'
```

Filtering data

- WHERE Clause
- Filtering with Text vs Numeric Fields
- Comparison operators (=, !=, >, <, >=, <=)
- BETWEEN
- IN
- LIKE
- Logic operators (AND, OR, NOT)
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```
1 SELECT * FROM Customers
2 WHERE Country='Germany' AND City='Berlin'
3
4
5 SELECT * FROM Customers
6 WHERE City='Berlin' OR City='München'
7
8
9 SELECT * FROM Products
10 WHERE Price NOT BETWEEN 10 AND 30
11
12
13 -- combine multi logic operator in one statment
14 SELECT * FROM Customers
15 WHERE Country='Germany' AND (City='Berlin' OR City='München')
16
```

Filtering data

- WHERE Clause
- Filtering with Text vs Numeric Fields
- Comparison operators (=, !=, >, <, >=, <=)
- BETWEEN
- IN
- LIKE
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```
1 SELECT CustomerName, ContactName, Address
2 FROM Customers
3 WHERE Address IS NULL
4
5
6 SELECT CustomerName, ContactName, Address
7 FROM Customers
8 WHERE Address IS NOT NULL
```

Agenda

- ⬡ What is RDBMS.
- ⬡ Tables, Columns and Data types
- ⬡ How to design a database.
- ⬡ One-To-Many & Many-To-Many Relationships.
- ⬡ Activity: FB, YouTube, Talabat
- ⬡ SQL
- ⬡ CRUD
- ⬡ Selecting data
- ⬡ Filtering data
- ⬡ **Ordering data**
- ⬡ Limiting data

- ⬡ Aggregate Functions
- ⬡ Grouping data
- ⬡ Datetime Operations
- ⬡ Joining tables
- ⬡ Subqueries
- ⬡ Window Functions
- ⬡ Inserting new data
- ⬡ Updating data
- ⬡ Deleting data
- ⬡ Python & MySQL
- ⬡ MySQL Workbench

Ordering data

Order rows by single column ascending

 Order rows by single column descending

 Filter then Order

 Order rows by several columns

```
1 SELECT * FROM Customers ORDER BY Country ASC
2
3 -- by default ORDER BY is in ascending order
4 SELECT * FROM Customers ORDER BY Country
5
```

Ordering data

- Order rows by single column ascending
- Order rows by single column descending
- Filter then Order
- Order rows by several columns

```
1 SELECT * FROM Customers ORDER BY Country DESC
```

Ordering data

- Order rows by single column ascending
- Order rows by single column descending
- Filter then Order**
- Order rows by several columns

```
1 SELECT * FROM Products
2 WHERE Price BETWEEN 10 and 30
3 ORDER BY ProductName DESC
```

Ordering data

- Order rows by single column ascending
- Order rows by single column descending
- Filter then Order
- Order rows by several columns

```
1 /*
2 Order first with Country ASC
3 and when there are many rows that have same countries
4 order them by CustomerName ASC
5 */
6 SELECT * FROM Customers
7 ORDER BY Country, CustomerName
8
9
10 /*
11 Order first with Country ASC
12 and when there are many rows that have same countries
13 order them by CustomerName DESC
14 */
15 SELECT * FROM Customers
16 ORDER BY Country ASC, CustomerName DESC
17
18
19 /*
20 Order first with Country DESC
21 and when there are many rows that have same countries
22 order them by CustomerName DESC
23 */
24 SELECT * FROM Customers
25 ORDER BY Country DESC, CustomerName DESC
```

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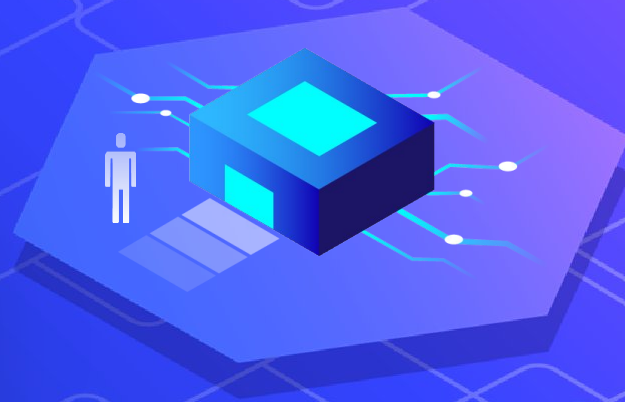
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Limiting data

- ⬡ Limit resulted rows to a specific number from first row
- ⬡ Limit resulted rows to a specific number from a specific offset row



```
1 -- return only the first 3 rows  
2 SELECT * FROM Customers LIMIT 3
```

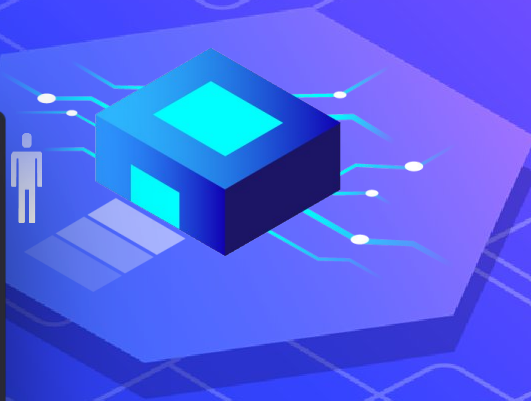


Limiting data

- Limit resulted rows to a specific number from first row
- Limit resulted rows to a specific number from a specific offset row



```
1 -- return only 3 rows after the second row
2 SELECT * FROM Customers LIMIT 2, 3
3
4 -- or you can use OFFSET keyword
5 SELECT * FROM Customers LIMIT 3 OFFSET 2
```



Questions ?!

